Data Structure Lab Assignment (CS 2172)
Assignment 3: Queue

Time: 2 weeks

1. Implementation of an Integer Queue

In this assignment you are required to implement a queue where integer data can be en-queued and de-queued. You should define (**typedef**) an appropriate type called ***queue*** such that multiple variables of type ***queue*** can be defined.

Your implementation should support the following functions (interface) for the queue.

1. ***Queue createIntegerQueue(int queueSize)*** - This allocates space for the queue to hold maximum "***queueSize***" number of integers and initializes that space. Its return type is "***queue***". The function returns ***NULL*** if creation of the queue fails.
2. ***int enqueueInteger(queue q, int d)*** - It en-queues the data ***d*** in the queue ***q***. It returns 1 if the operation is successful. If the operation fails (say, when queue ***q*** is full and ***d*** cannot be en-queued), the function returns 0.
3. ***int dequeueInteger(queue q, int *dp)*** - It de-queues from the queue ***q*** and stores the dequeued element at address ***dp***. It returns 1 if the operation is successful. If the operation fails (say, when queue s is empty and ***dequeueInteger()*** is attempted), the function returns 0.
4. ***int freeIntegerQueue(queue q)*** - It frees the space allocated for queue ***q***. It returns 1 if the operation is successful. If the operation fails (say, ***q*** does not refer to a valid queue), the function returns 0.
5. ***int isIntegerQueueFull(queue q)*** - It returns 1 if the queue associated with ***q*** is full. The function returns 0 otherwise. If ***q*** does not refer to a valid queue then too the function returns 1.
6. ***int isIntegerQueueEmpty(queue s)*** - It returns 1 if the queue associated with ***q*** is empty. The function returns 0 otherwise. If ***q*** does not refer to a valid queue then too the function returns 1.

Note: You should implement queue functionality using the circular queue concept. This will be done in two stages

Stage-1: Initially avoid having a ***count*** variable that represents the number of elements in the queue. This is for better understanding of the difficulties,.

Stage-2: Then finally, introduce the ***count*** variable to make it simple.

Write suitable main() function to demonstrate that your functions are working as desired.

2. Using above Queue implementation, simulate the following

Assume there is one queue created as ***myQueue*** of size ***N***. Also, consider that a series of **positive** integers are read from the user and queued into ***myQueue***. The process of populating into the queue continues till the **queue overflows**. Now perform the following steps of operation on each element of the queue.

<table>
<tr><td>Service operation detail</td><td>Diagram</td></tr>
<tr><td>

Dequeue an element from myQueue. Let's call that element as qElement.
Process the qElement element as follows

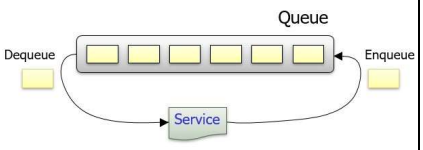
✓ qElement = qElement – rValue; where rValue is a fixed number, let's say 3.

If updated qElement is positive

✓ Enqueue back the updated qElement into myQueue.

Else

✓ Drop the qElement

</td><td> 
</td></tr>
<tr><td colspan="2">This process continues till the time queue underflows, such that no element present in the queue.</td></tr>
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Advancement: Modify the problem where ***rValue*** is a randomly generated positive integer between (1 – 9) and repeat testing.

Note: Please ask to the instructor for more testing and observations of this problem.